

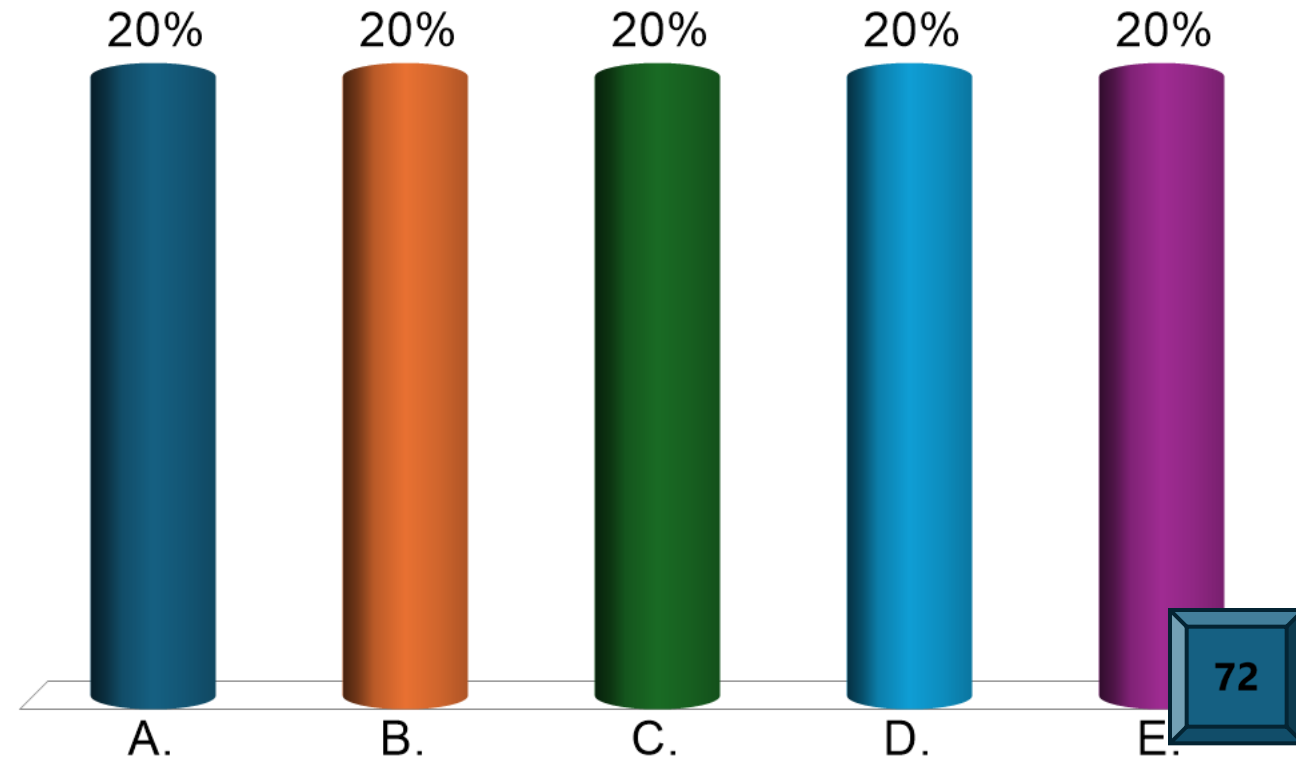


BPM 1
CPR IMCQ 5

KEY
Fall 2025

1. A 6-year-old boy is brought to the physician because of recurrent respiratory and gastrointestinal infections since infancy. His medical history is notable for multiple episodes of otitis media, pneumonia, and chronic diarrhea. Physical examination shows small tonsils and absent palpable lymph nodes. Laboratory studies show markedly decreased levels of immunoglobulins. Which of the following best characterizes the type of immune response most likely affected in this case?

- A. Rapid
- B. Non-specific
- C. Specific
- D. Does not generate memory
- E. Inflammation

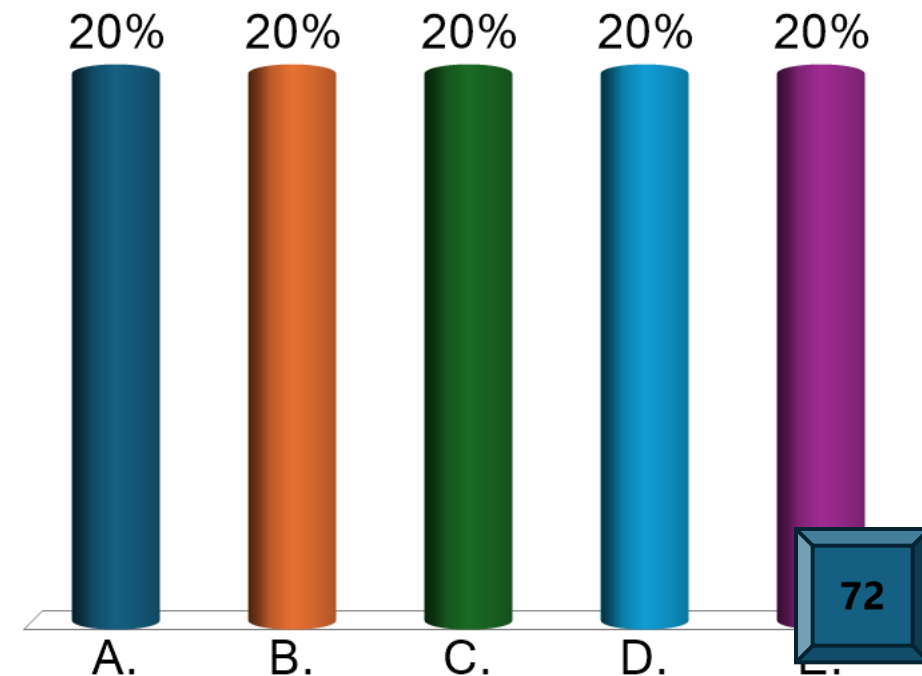


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2. A 3-month-old boy is being evaluated for recurrent infections. Since birth, he has had persistent oral thrush, several episodes of viral pneumonia, and chronic diarrhea. Physical examination shows facial dysmorphism with low-set ears, micrognathia, and a small mandible. Cardiac auscultation shows a holosystolic murmur. A chest X-ray shows an absence of a thymic shadow. Laboratory studies show a lymphopenia. Flow cytometry shows absent CD3+ cells and normal CD19+ cells. Which of the following best identifies the function of the cell type most likely affected?

- A. Differentiates to plasma cells
- B. Facilitates the humoral immune response
- C. Facilitates the cell-mediated immune response
- D. Production of antibodies
- E. Phagocytosis at the site of infections



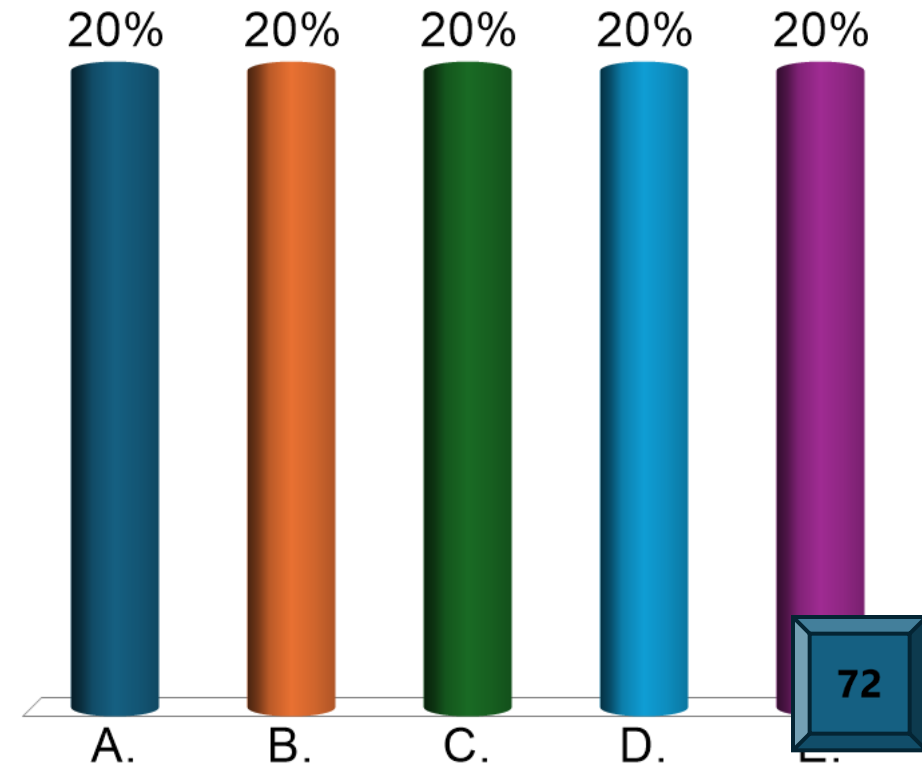
2. A 3-month-old boy is being evaluated for recurrent infections. Since birth, he has had persistent oral thrush, several episodes of viral pneumonia, and chronic diarrhea. Physical examination shows facial dysmorphism with low-set ears, micrognathia, and a small mandible. Cardiac auscultation shows a holosystolic murmur. A chest X-ray shows an absence of a thymic shadow. Laboratory studies show a lymphopenia. Flow cytometry shows absent CD3+ cells and normal CD19+ cells. Which of the following best identifies the function of the cell type most likely affected?

SOM.MK.1.BPM1.3.CPR.1.HCB.1214 Identify the function(s) of the following blood cells: CD4+ & CD8+ T cells, B cells & plasma cells and natural killer cells.

- A. Differentiates to plasma cells
- B. Facilitates the humoral immune response
- C. Facilitates the cell-mediated immune response
- D. Production of antibodies
- E. Phagocytosis at the site of infections

3. A 7-month-old girl is brought to the physician by her parents because of a 5-month history of recurrent bacterial and viral infections and failure to thrive. Genetic testing shows a defect in the expression of MHC-II molecules on the surface of some immune-fighting cells. Which of the following cell types is most likely affected?

- A. T-cell/T-lymphocytes
- B. Neutrophils
- C. Antigen presenting cells
- D. Monocytes
- E. Platelets



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- A. T-cell/T-lymphocytes
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- E. Platelets

4. A 23-year-old African American man comes to the emergency department because of sudden-onset fatigue, dark urine, and back pain that began earlier today. Two days ago, he developed dysuria and self-treated with leftover trimethoprim-sulfamethoxazole. His temperature is 37.1°C (98.8°F), blood pressure is 124/76 mm Hg, and pulse is 96/min. Physical examination shows scleral icterus, mild right upper quadrant tenderness and the spleen tip is palpable. Results of laboratory studies are shown. Urinalysis shows numerous RBCs and hemoglobin. Which of the following best explains the patient's clinical presentation?

Laboratory test	Patient value	Normal range (adult male)
Hemoglobin	8.9 g/dL	13.5-17.5 g/dL
Reticulocyte count	Increased	0.5-2%
Lactate dehydrogenase	Increased	140–280 U/L
Haptoglobin	Decreased	30-200 mg/dL
Peripheral smear	Bite cells, Heinz body changes (after staining)	

- A. Increased production of NADPH 20%
- B. Increased activity of glutathione reductase 20%
- C. Decreased availability of reduced glutathione 20%
- D. Autoantibody-mediated destruction of RBCs 20%
- E. Inhibition of superoxide dismutase 20%

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SOM.MK.I.BPM1.1.CPR.2.BCHM. 0268 Evaluate the use of NADPH in: Detoxification of hydrogen peroxide, Phagocytosis, Nitric oxide synthesis, Reductive biosynthesis

5. A 13-year-old girl is brought to the physician by her mother because of a 6-month history of axillary and inguinal swellings and weight loss. Physical examination shows palpable, non-tender enlargement of cervical, axillary, and inguinal lymph nodes. A lymph-node biopsy shows granulomas with central necrosis and fungi, which are identified as *Paracoccidioides* spp. She is subsequently referred for assessment for an immunodeficiency disorder. Immunocytochemistry shows very low neutrophil myeloperoxidase activity. Formation of which of the following intermediates is most likely decreased in this patient?

- | | |
|--------------------------------------|-----|
| A. Superoxide (O_2^-) | 14% |
| B. NADPH | 14% |
| C. Hydroxyl radical ($\bullet OH$) | 14% |
| D. Hypochlorous acid (HOCl) | 14% |
| E. Hydrogen peroxide (H_2O_2) | 14% |
| F. Reduced glutathione (GSH) | 14% |
| G. Nitric oxide (NO) | 14% |

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- F. Reduced glutathione (GSH)
- G. Nitric oxide (NO)

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0279; Predict the effect of NADPH oxidase deficiency and myeloperoxidase deficiency. Indicate the biochemical defect in chronic granulomatous disease. CPR DLA 10: SOM.MK.I.BPM1.1.CPR.2.BCHM. 0277; Discuss the beneficial role of ROS formation as an immunological response in phagocytosis and describe the reactions catalyzed by NADPH oxidase and myeloperoxidase.

6. A 34-year-old man comes to the clinic because of a 3-month history of progressive numbness and burning in the feet and weakness in the legs. His diet consists mainly of polished white rice. Physical examination shows decreased ankle reflexes and bilateral distal lower extremity weakness. Laboratory testing shows low erythrocyte transketolase activity. Which of the following vitamins is most likely deficient in this patient?

- | | |
|---------------|-----|
| A. Cobalamin | 14% |
| B. Niacin | 14% |
| C. Pyridoxine | 14% |
| D. Thiamine | 14% |
| E. Riboflavin | 14% |
| F. Folic acid | 14% |
| G. Vitamin C | 14% |

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C. Pyridoxine

✓ D. Thiamine

E. Riboflavin

F. Folic acid

G. Vitamin C

SOM.MK.I.BPM1.1.CPR.2.BCHM.0266 Provide an overview of the intermediates of the non-oxidative phase of the PPP highlighting the importance of transketolase.

7. A 4-year-old boy is brought to the clinic by his parents for a follow-up examination. He has a 2-year history of multiple episodes of pneumonia, skin abscesses, and lymphadenitis. Infections have often been caused by *Candida albicans* and *Serratia marcescens*. Physical examination shows cervical lymphadenopathy and several healed scars from prior skin infections. A neutrophil function test shows a negative dihydrorhodamine test, indicating no increase in oxygen utilization during phagocytosis. Which of the following is the most likely diagnosis?

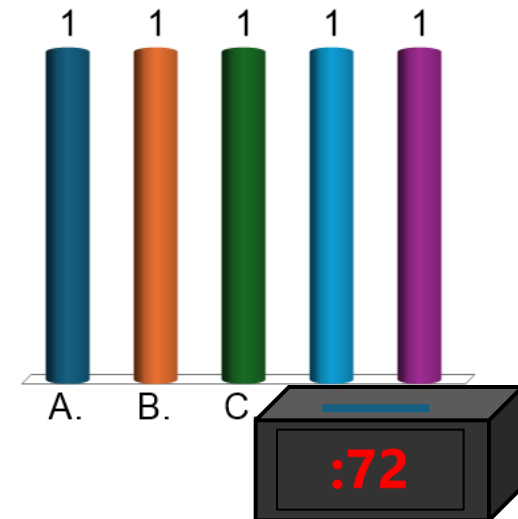
- | | |
|---|-----|
| A. Nitric oxide synthase deficiency | 20% |
| B. Chronic granulomatous disease | 20% |
| C. Myeloperoxidase deficiency | 20% |
| D. Amyotrophic lateral sclerosis | 20% |
| E. Glucose 6-phosphate dehydrogenase deficiency | 20% |

7. A 4-year-old boy is brought to the clinic by his parents for a follow-up examination. He has a 2-year history of multiple episodes of pneumonia, skin abscesses, and lymphadenitis. Infections have often been caused by *Candida albicans* and *Serratia marcescens*. Physical examination shows cervical lymphadenopathy and several healed scars from prior skin infections. A neutrophil function test shows a negative dihydrorhodamine test, indicating no increase in oxygen utilization during phagocytosis. Which of the following is the most likely diagnosis?

- A. Nitric oxide synthase deficiency
- ✓ B. Chronic granulomatous disease
- C. Myeloperoxidase deficiency
- D. Amyotrophic lateral sclerosis
- E. Glucose 6-phosphate dehydrogenase deficiency

8. A 41-year-old woman comes to the physician because of a 6-week history of recurrent episodes of headaches, shortness of breath on exertion, and pain in the left loin. She has a family history of kidney disease. Ten months ago, her sister died due to a brain hemorrhage associated with a kidney disease. Her blood pressure is 164/98 mm Hg; other vital signs are within normal limits. Physical examination shows a palpable mass in the left costovertebral angle posteriorly. Cardiac auscultation shows a loud S2 in the aortic area. Abdominal ultrasound shows an enlarged left kidney with multiple cysts. She is diagnosed with polycystic kidney disease. Serum studies show elevated renin concentration. Which of the following sets of serum alterations is most likely in this patient?

Option	Angiotensin II	Aldosterone	Epinephrine and norepinephrine
A	No change	Increased	Increased
B	Increased	Increased	Increased
C	Increased	Increased	No change
D	Decreased	Decreased	No change
E	No change	No change	Increased



8. A 41-year-old woman comes to the physician because of a 6-week history of recurrent episodes of headaches, shortness of breath on exertion, and pain in the left loin. She has a family history of kidney disease. Ten months ago, her sister died due to a brain hemorrhage associated with a kidney disease. Her blood pressure is 164/98 mm Hg; other vital signs are within normal limits. Physical examination shows a palpable mass in the left costovertebral angle posteriorly. Cardiac auscultation shows a loud S2 in the aortic area. Abdominal ultrasound shows an enlarged left kidney with multiple cysts. She is diagnosed with polycystic kidney disease. Serum studies show elevated renin concentration. Which of the following sets of serum alterations is most likely in this patient?

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A	No change	Increased	Increased
B	Increased	Increased	Increased
✓ C	Increased	Increased	No change
D	Decreased	Decreased	No change
E	No change	No change	Increased

SOM.MK.II.BPM1.3.CPR.2.PHYS. 0179. Define and classify hypertension.

9. A 35-year-old man is brought to the emergency department because of a 1-hour history of severe headache, blurred vision, and muscle weakness. He has a 2-year history of chronic kidney disease. His blood pressure is 210/109 mm Hg. Physical examination shows muscle weakness in the left upper and lower limbs. Funduscopic examination shows papilledema and retinal hemorrhages. He is diagnosed with hypertensive emergency and treated with intravenous infusion of nitroprusside, a drug that causes arteriolar and venous dilation. His blood pressure drops to 142/84 mm Hg, and his condition improves. Which of the following sets of cardiovascular alterations is most likely in this patient after the treatment?

Option	Blood volume	Total peripheral resistance	Afterload on left ventricle
A	No change	Decreased	Increased
B	Increased	Increased	Decreased
C	Decreased	Decreased	Increased
D	Decreased	Increased	No change
E	No change	Decreased	Decreased



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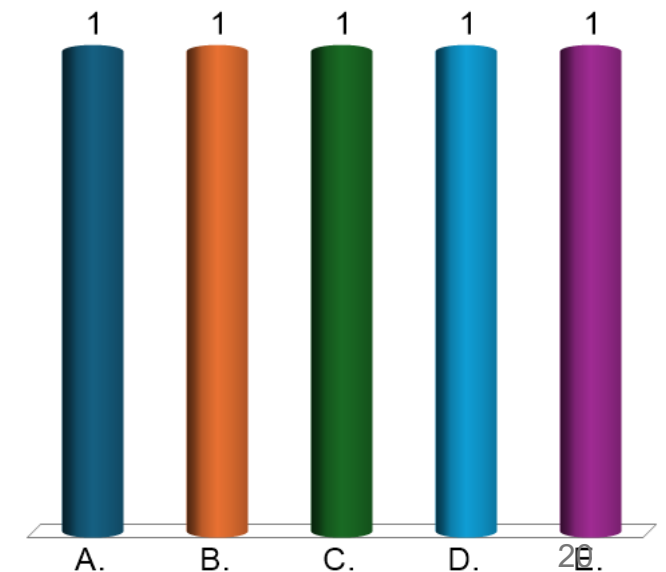
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A	No change	Decreased	Decreased
B	Increased	Increased	Decreased
C	Decreased	Decreased	Increased
D	Decreased	Increased	No change
✓ E	No change	Decreased	Decreased

SOM.MK.II.BPM1.3.CPR.2.PHYS.01
81. Describe the compensatory responses to mild cardiac failure and explain the outcomes of moderate to severe cardiac failure.



10. A 37-year-old woman comes to the clinic because of a 10-month history of progressive shortness of breath and fatigue. She has no major chronic illness. Her temperature is 37°C (98.6°F), pulse is 76/min, blood pressure is 121/77 mmHg, and oxygen saturation is 89% on room air. Physical examination shows a loud S2 in the pulmonary area. A CT scan of the thorax shows normal lung parenchyma. Lung function studies show normal lung volumes and capacities, but elevated mean pulmonary arterial pressure and reduced diffusing capacity of lung. Echocardiography shows right ventricular hypertrophy but a normal left ventricle. She is diagnosed with pulmonary hypertension. Increase in which of the following best explains the echocardiographic findings of this patient?

- A. Afterload on the left ventricle
- B. Preload on the left ventricle
- C. Total systemic vascular resistance
- D. Afterload on the right ventricle
- E. Preload on the right ventricle



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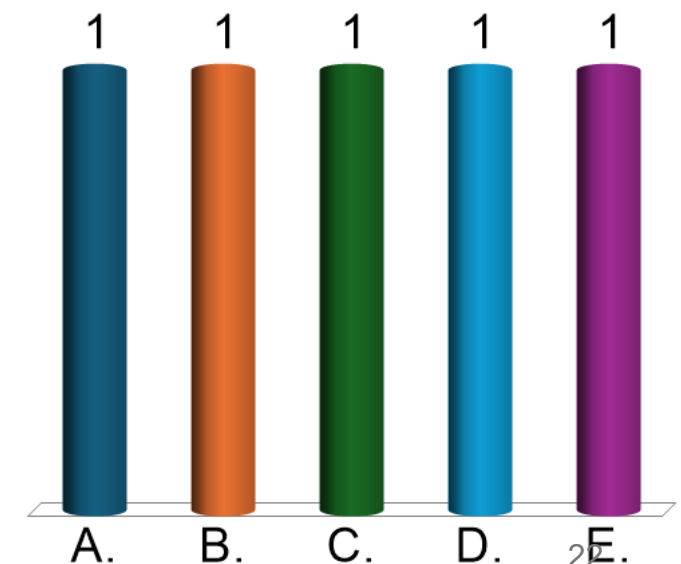
- A. Afterload on the left ventricle
- B. Preload on the left ventricle
- C. Total systemic vascular resistance
- ✓ D. Afterload on the right ventricle
- E. Preload on the right ventricle

SOM.MK.II.BPM1.3.CPR.2.PHYS.
0179. Define and classify
hypertension.



11. A 39-year-old woman comes to the physician because of a 6-month history of recurrent episodes of headache, shortness of breath on exertion, and muscle weakness. She does not smoke cigarettes, drink alcoholic beverages, or use illicit drugs. Her blood pressure is 182/98 mm Hg; other vital signs are within normal limits. Physical examination shows a loud S2 in the aortic area. An ECG shows left axis deviation. Laboratory studies show elevated serum aldosterone concentration but reduced serum renin concentration. Abdominal ultrasonography shows a 2-cm mass in the left adrenal cortex. She is diagnosed with aldosterone-secreting tumor. Which of the following mechanisms best explains the cause of hypertension in this patient?

- A. Increased viscosity of blood due to increased red cell volume
- B. Increased sodium and water retention
- C. Increased total peripheral resistance
- D. Increased arteriolar tone
- E. Decreased elasticity of the arteries



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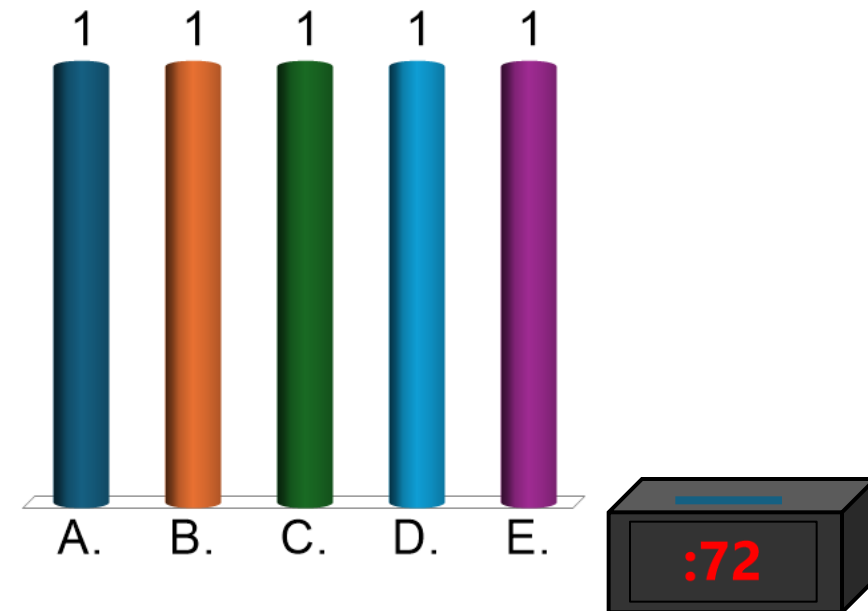
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- ✓ B. Increased sodium and water retention
- C. Increased total peripheral resistance
- D. Increased arteriolar tone
- E. Decreased elasticity of the arteries

SOM.MK.II.BPM1.3.CPR.2.PHYS.0341. Discuss long-term regulation of blood pressure by Renin-angiotensin-aldosterone mechanism
SOM.MK.II.BPM1.3.CPR.2.PHYS.0179. Define and classify hypertension.



12. A 57-year-old man comes to the physician because of a 4-week history of shortness of breath at night and swelling of the feet. One year ago, he was diagnosed with a mediastinal lymphoma and has had five cycles of radiation to the mediastinum so far. His pulse is 86/min, respirations are 20/min, and blood pressure is 110/72 mm Hg. Physical examination shows distended neck veins and pitting edema of the feet. Coarse crackles are heard over the lower lung fields bilaterally. Cardiac auscultation shows a loud S4 in the mitral area. Echocardiography shows reduced end-diastolic volume, a stiff left ventricle, but normal ejection fraction. He is diagnosed with heart failure due to diastolic dysfunction. Which of the following cardiovascular parameters is most likely increased in this patient?

- A. Stroke volume
- B. Mean aortic pressure
- C. End systolic volume
- D. Left ventricular compliance
- E. End diastolic ventricular pressure



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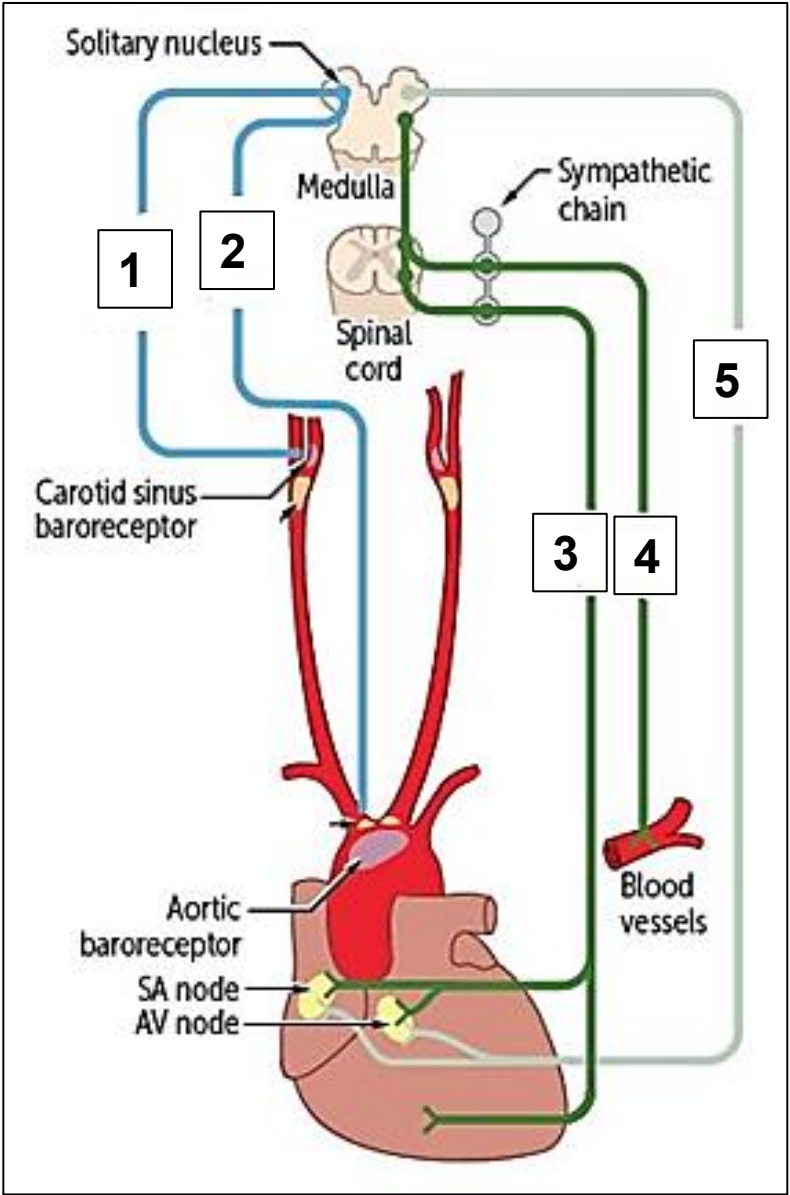
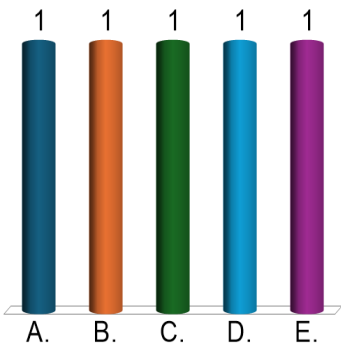
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SOM.MK.II.BPM1.3.CPR.2.PHYS.0181. Describe the compensatory responses to mild cardiac failure and explain the outcomes of moderate to severe cardiac failure.



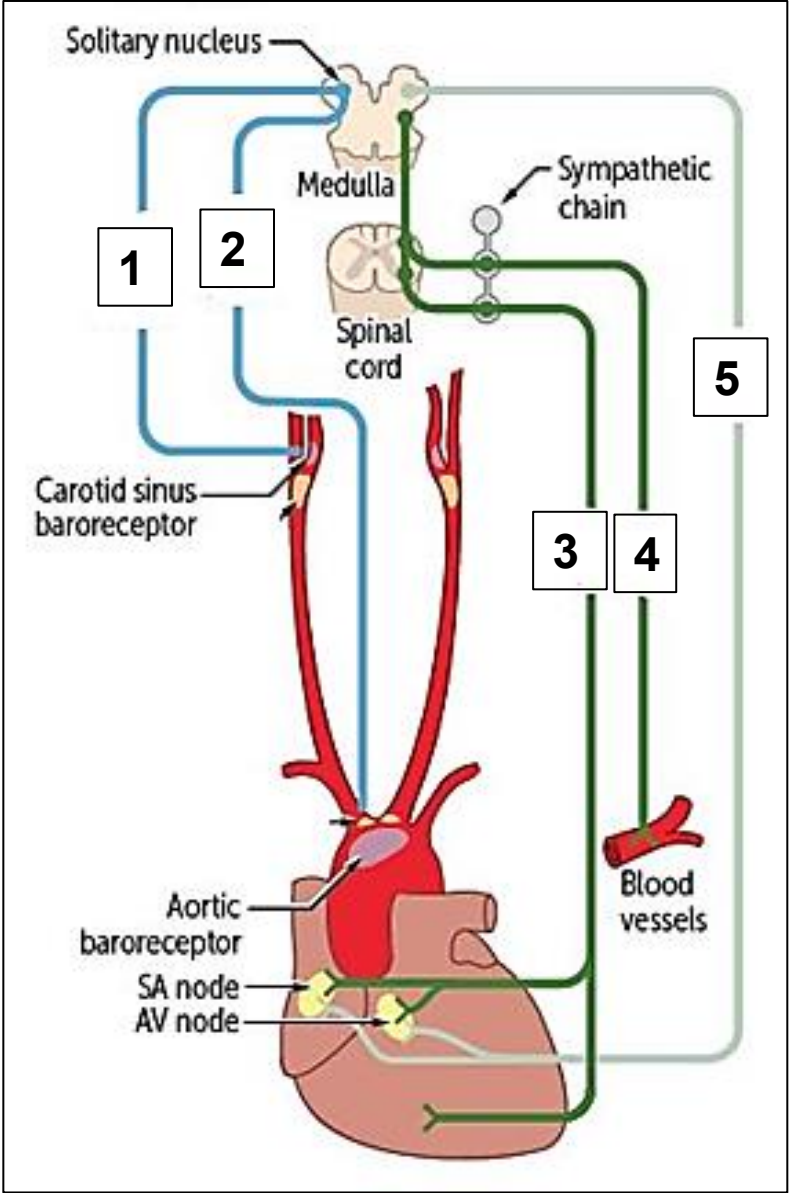
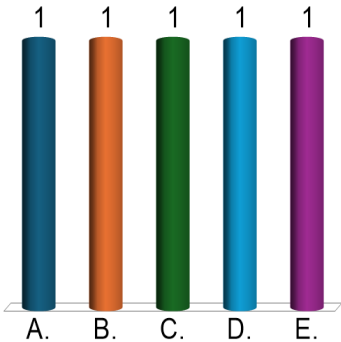
13. Investigators conduct a study related to the effect of stimulation of baroreceptors on blood pressure. The nerve labeled 1 in the figure is electrically stimulated for 10 seconds and its effect on pulse, heart rate, blood pressure, cardiac output, venous return, and serum catecholamine concentration is recorded. Which of the following sets of alterations is most likely observed in this study?

Option	Pulse rate	Cardiac output	Vagal tone
A	↑	↑	↑
B	↓	↓	↓
C	↓	↓	↑
D	↓	↑	↑
E	↑	↓	↑



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C	↓	↓	↑
D	↓	↑	↑
E	↑	↓	↑

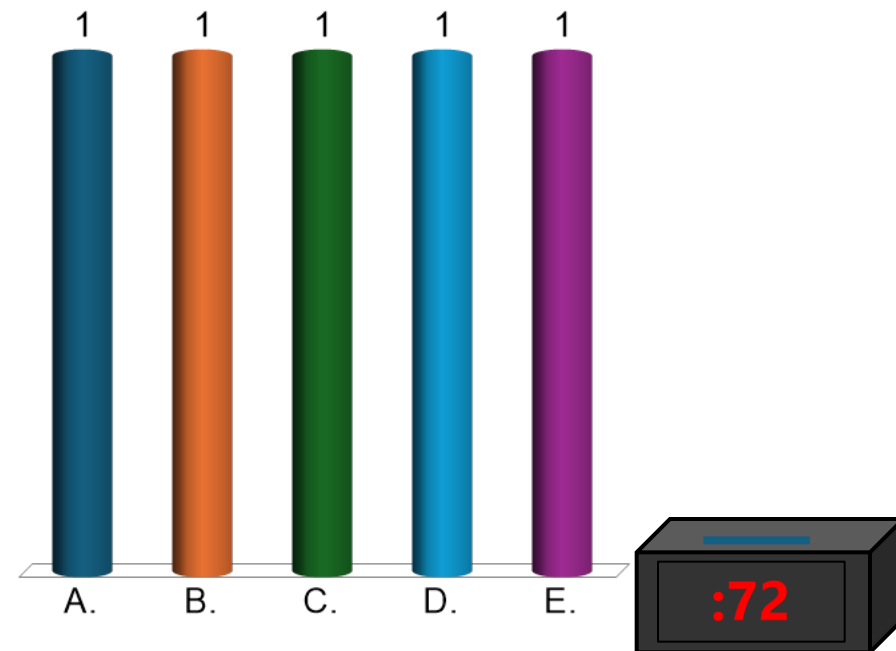


SOM.MK.II.BPM1.3.CPR.2.PHYS.0340. Construct a step-by-step flow chart to indicate changes in the cardiovascular system that occur when the blood pressure changes in the body, taking examples of increase and decrease in blood pressure.



14. A 78-year-old man comes to the physician because of a 5-day history of progressive shortness of breath, swollen feet, pain in the right upper abdomen, and fatigue. He has a 30-year history of chronic obstructive lung disease (COPD). Vital signs are within normal limits. Physical examination shows distended neck vein, elevated JVP, positive hepatojugular reflux, and tender hepatomegaly. Pulmonary examination shows a hyperinflated chest, prolonged expiration, but no basal rales. Cardiac auscultation shows a loud S2 in the pulmonary area. Echocardiography shows right ventricular hypertrophy. He is diagnosed with right heart failure due to COPD. Which of the following additional clinical features is most likely in this patient?

- A. Peripheral edema
- B. Pulmonary edema
- C. Systemic hypertension
- D. Increased pulse pressure
- E. Absent “a” waves in JVP



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SOM.MK.II.BPM1.3.CPR.2.PHYS.0181. Describe the compensatory responses to mild cardiac failure and explain the outcomes of moderate to severe cardiac failure

15. A 28-year-old woman is brought to the emergency department 35 minutes after she was involved in a motor vehicle collision. She has multiple fractures, altered mental status, and is bleeding from the open wounds. She is estimated to have lost about 15% blood volume. Her pulse is 126/min and blood pressure is 92/64 mm Hg. Physical examination shows a low volume pulse, diaphoresis, and cold extremities. She is given supportive treatment and intravenous fluids. Which of the following sets of alterations is most likely when this patient is brought to the emergency department?

Option	Action potentials in sinus nerve	Action potentials in vagus nerve	PCWP	
A	Decreased	Decreased	Decreased	1
B	Increased	Increased	Increased	1
C	Decreased	Increased	Increased	1
D	Decreased	Increased	Decreased	1
E	Decreased	Decreased	Increased	1

PCWP = Pulmonary capillary wedge pressure



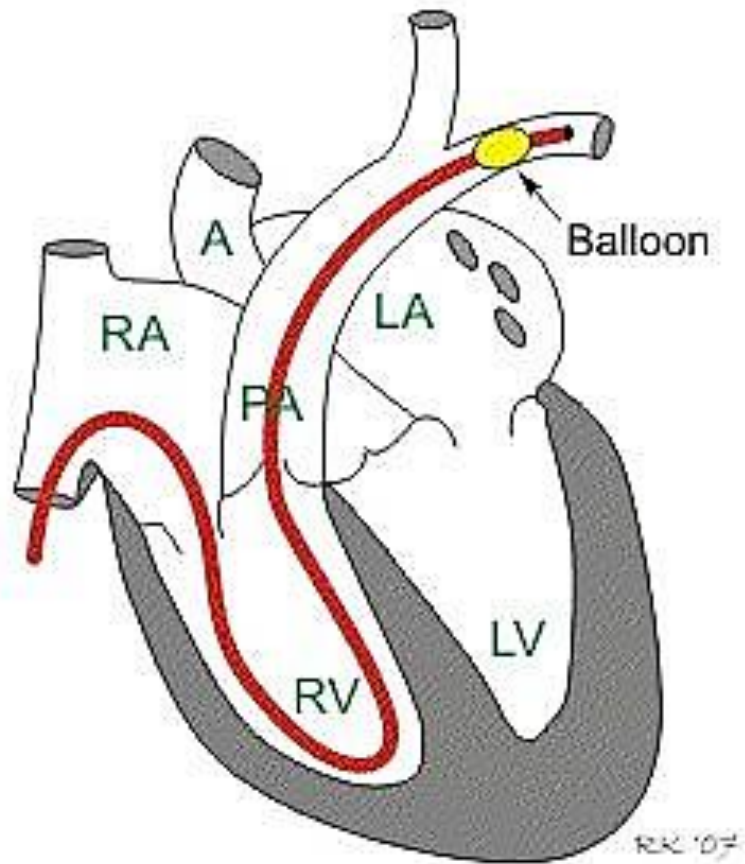
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B	Increased	Increased	Increased
C	Decreased	Increased	Increased
D	Decreased	Increased	Decreased
E	Decreased	Decreased	Increased

SOM.MK.II.BPM1.3.CPR.2.PHYS.01
83. Describe the direct cardiovascular consequences of the loss of 30% of the circulating blood volume on cardiac output, central venous pressure, and arterial pressure.

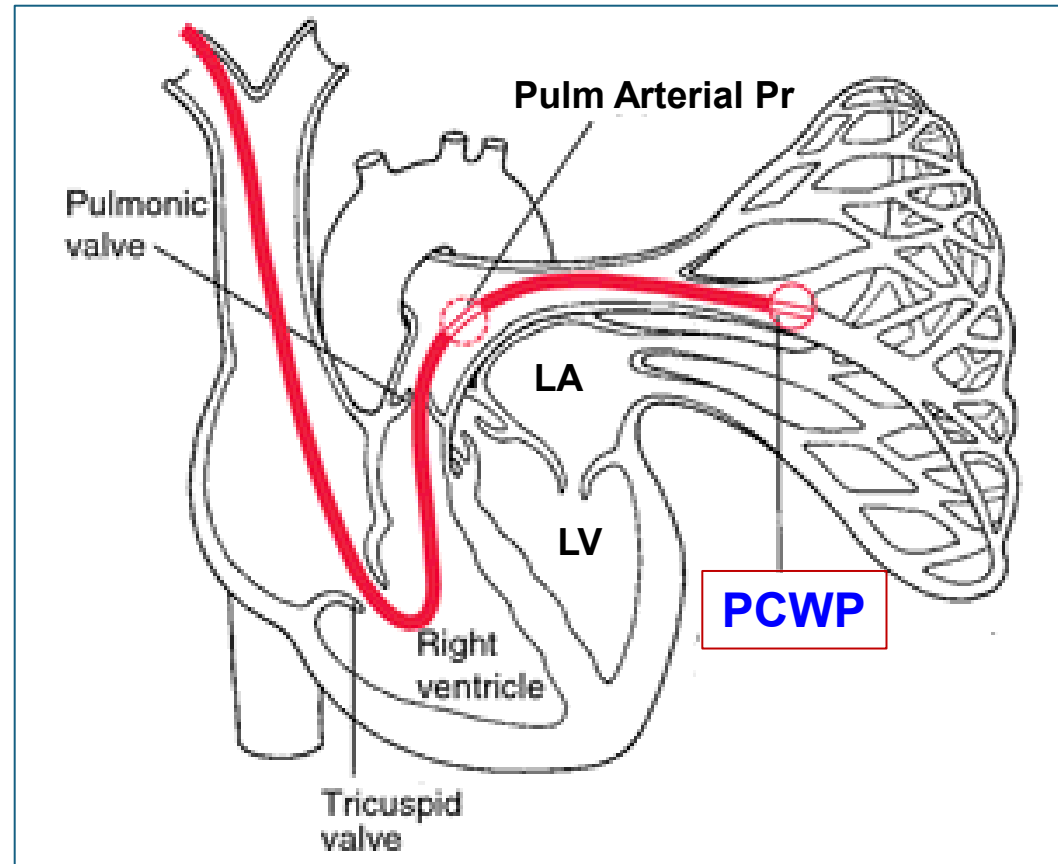
PCWP = Pulmonary capillary wedge pressure





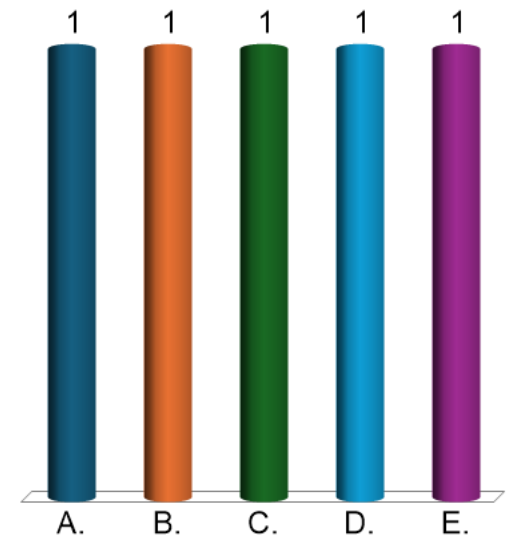
Balloon-tipped, Swan-Ganz catheter for measuring pulmonary capillary wedge pressure (PCWP).

PCWP measures Left atrial pressure (LAP)



16. A 42-year-old man is brought to the emergency department after being rescued from a snowstorm in which he was stranded in a valley for 20 hours. Three days ago, he went on a hike and lost communication with his family. His temperature is 34.5°C (94°F), pulse is 48/min, blood pressure is 110/64 mm Hg, and respirations are 12/min. Physical examination shows lethargy, impaired judgement, hyporeflexia, and cold extremities, nose and earlobes. Pupillary light reflexes are sluggish. Which of the following mechanisms best explains the changes in this patient's extremities, nose and earlobes?

- A. Increased heat loss from skin
- B. Activation of sympathetic cholinergic nerves to skin
- C. Closure of arteriovenous anastomoses
- D. Inhibition of sympathetic adrenergic nerves to skin
- E. Autoregulation of skin blood flow

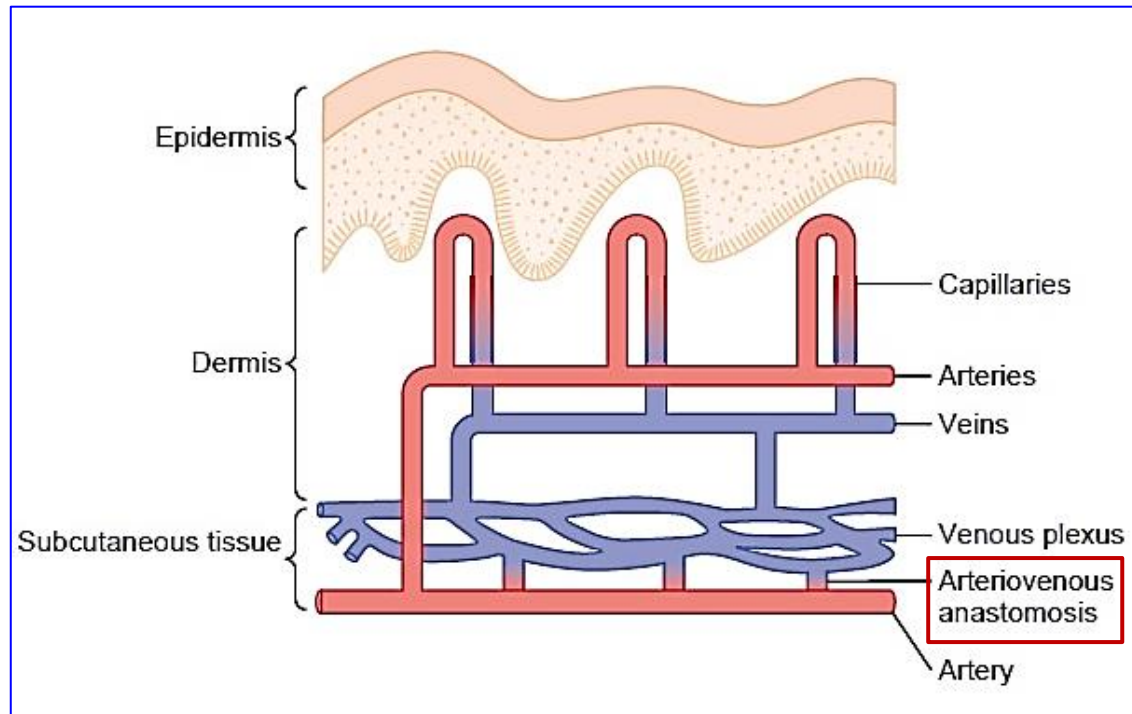


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Adjustment of cutaneous blood flow is used for body temperature regulation: Role of Arteriovenous anastomoses (AVA)

- Direct vascular connections between arteries and veins exist in the skin. These are known as **Arteriovenous anastomoses (AVA)**. These are richly innervated by sympathetic nerves.
- AVA are densely present in skin over the **hands, feet, ears, nose, and face**. They are less in skin over the limbs and torso.
- These AVA function as shunts: When AVA open, allow blood to flow directly into a large surface area of small venous vessels (venous plexus shown in the figure) just underlying the surface of the skin. Blood flow to skin decreases when these AVA are closed.



Both **sympathetic adrenergic** and **sympathetic cholinergic** nerves innervate arteriovenous anastomoses.

Adrenergic n activity: closes AVA;
Cholinergic n activity opens AVA.

Note: Adrenergic nerve releases Norepinephrine and cholinergic nerve releases Acetylcholine as neurotransmitters.

You've worked hard for this
exam, so give your best effort!

**Best Wishes
To You.**

